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TCS–410

B. TECH. (ECE)

(FOURTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

DATA STRUCTURE USING C

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Find the time complexity of the following code : (CO1, CO2)

```
for (i=1;i<=m;i=i×2)
  for (j=1;j<=i;j++)
  printf("%d%d",i,j);
```

P. T. O.

OR

(b) In a two dimensional matrix A[0,...,r – 1, 0...,c – 1] the location of A[i, j] is same in both row- major-order and column-major-order. Find the relationship between the number of rows and number of columns (r, c) with the cell index (i, j) and justify the relation with one example.(CO1, CO2)

2. (a) Using malloc() write the code section to do the memory allocation for one matrix of size M × N. (CO1, CO2)

OR

(b) Write algorithm or pseudo code for deleting all the nodes from the single linked list which are divisible by a given number k. (CO1, CO2)

3. (a) Write a recursive function to display elements of stack using the operation push() and. (CO1,CO2)

OR

(b) Given two sorted arrays A and B. Array is full and array B is partially full and the number or empty slots are just enough to accommodate all elements of array A. Write one C function to merge the two soted arrays to fill the array B without considering the third array. (CO1, CO2)

4. (a) Write a C function which will take one unsorted linked list with redundant elements and will output one sorted linked list with unique elements. (CO1, CO2)

OR

(b) What are the different way of representing sparse matrix in memory. Explain through example. Using any suitable representation write the C-function to multiply two sparse matrices. (CO1, CO2)

5. (a) Write a C-function to reverse the nodes of a double linked list and justify its correctness with suitable example.

(CO1, CO2)

OR

(b) What are the different data structure used to represent the stack ADT ? Write the Push() operation using one data structure and Pop() operation using the other data structure of stack ADT. (CO1, CO2)